ThreadUD – Mobile App

Peer-Learning Forum for TU Dublin Students

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1. Introduction

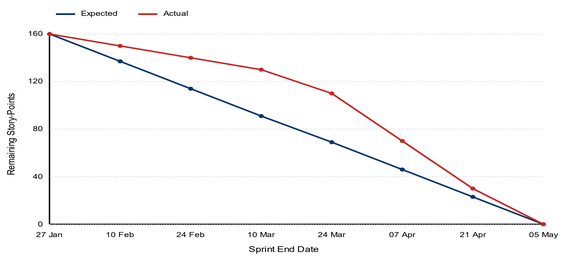
Lecture time is limited, and Stack Overflow isn't course-specific. ThreadUD fills that gap with a real-time, student-only discussion space, letting classmates crowd-source answers, share resources and stay engaged beyond the classroom.

2. Objective

- **Q&A Forum** – module-filtered threads with nested posts
- **Immediate feedback** – live comments & replies
- **Safe space** – AI moderator + toxicity filter
- **Seamless UX** – smooth UI on any Android phone and easy to use

3. Methodology

Agile Scrum (2-week sprints, weekly meetings) kept scope tight while user testing usability after each new implementation. CI/CD via GitHub commits and updates delivered rapid releases .



4. Technologies

React Native & Expo	Node.js & Express.js
TypeScript & JavaScript	MongoDB & Mongoose
TailwindCSS/NativeWind	Bycrypt and JWT
AsyncStorage	Nodemailer
Socket.io Client	Google Perspective API

5. Features & Functionality

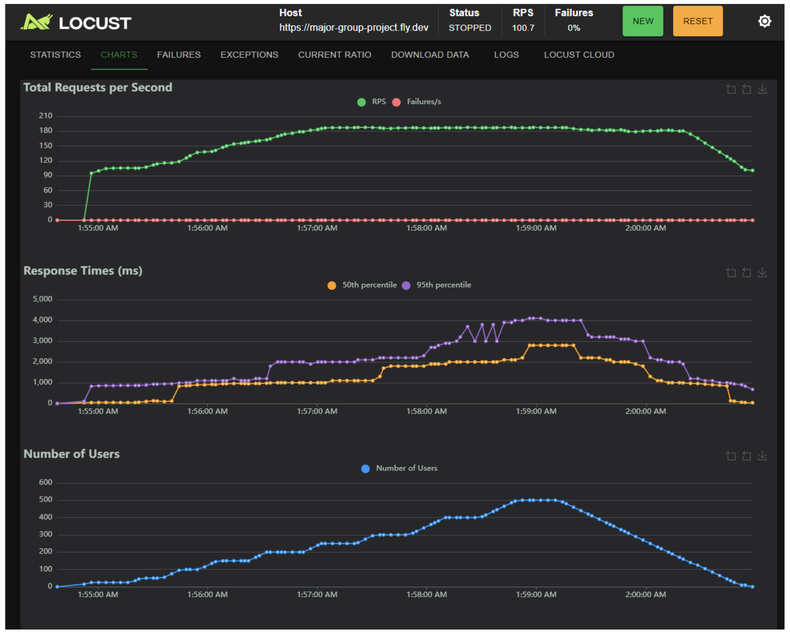
- Personalized home feed (joined threads)
- Thread page join/leave and view & interact with posts
- Post Page comments & nested replies
- Search Page by Thread name/keywords/Course code
- Make a Post under a Thread you are a part of
- Make a Thread for any course in any year of study
- Profile Page pulls user info/posts/threads and can update profile change password/course/year of study
- Login/Sign-up with authentication and forgot password
- Admin dashboard for flag review posts & comments

6. Security & Moderation

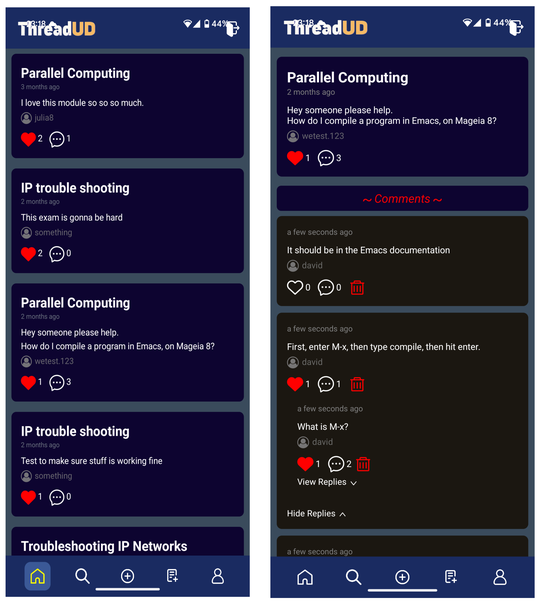
- Bcrypt-hashed passwords
- HTTPS-only traffic
- JWT refresh tokens secure accounts
- AI moderator pre-screens every post flagging the bad ones based on a custom scoring system
- Flagged items (do not go public) enter an admin review queue, keeping discourse clean and civil

7. Testing

- User tests done on college open day
- Stress testing on server done with Locust
- User interface testing
- Requirements and Use Case testing



8. Interface Design



9. Conclusions & Future Work

ThreadUD shows a robust mobile peer forums application that can help education and build community among students. In future we may look at adding notifications, rich media attachments and sharing posts.
This was a project we truly enjoyed working on and developing this vision we had in Oct.

REFERENCES

React Native: https://reactnative.dev/
Stack Overflow: https://stackoverflow.com/
Expo: https://expo.dev/
Locust: https://docs.locust.io/en/stable/